

Patient-Friendly Radiology Reporting—Implementation and Outcomes

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Abstract

Objective: To evaluate patient use and experience with patient-centered radiology reports provided via a radiology-specific patient portal in an outpatient setting.

Methods: Patient-centered interactive radiology reports were made available to all patients who had imaging examinations performed at any of seven outpatient imaging centers via a radiology-specific patient portal, using commercially available software. We recorded portal usage statistics including report access rate, report view time, and use of interactive diagrams and plain language explanations. We assessed patients' subjective report comprehension and experience via survey questions. Results were stratified by age.

Results: The majority of patients accessed their online radiology report (59%). For patients who used the interactive plain language definitions and diagrams, the average report viewing time was 5.8 ± 6.6 min compared with 1.8 ± 1.9 min for patients viewing only the raw text report ($P < .005$). The majority (84.7%) of patients reported that the definitions and diagrams helped them understand their report, and 86.7% of patients endorsed an overall positive experience viewing their report online.

Conclusions: Our results suggest that patient-centered radiology reporting is feasible in the outpatient private practice setting and that dedicated patient-centered content has the potential to increase engagement across all age groups. Our experience suggests that patient-centered radiology reports may improve subjective patient comprehension and positively impact the patient experience.

Key Words: 21st Century CURES Act, health IT, patient-centered reporting, patient engagement, plain language medical reports

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INTRODUCTION

Patients have been granted unprecedented access to their health records, including radiology results, through online patient portals [1]. More than 90% of patients now have access to their medical records through an online portal

[1]. Portal use is on the rise, with 28% of patients self-reporting portal use in 2017 compared with 14% in 2014 [2]. In a recent survey, most academic medical centers reported between 25% and 50% portal use [3]. One study found that 51% of patients within a single large health system viewed their radiology report in the online portal, and radiology reports were the second most frequently viewed record after lab results [4]. Recent federal legislation may increase these numbers further. In 2016, the US Congress passed the 21st Century Cures Act, which includes an information-blocking provision requiring that patients be granted access to all health information as soon as the information is available in the electronic health record [5,6].

Immediate radiology report access has the potential to benefit patients, because studies have shown that patients who understand their disease processes are more likely to adhere to treatment plans and have better outcomes [2,7,8]. However, radiology reports are technical and often jargon-laden documents intended for other physicians. There is

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evidence that patients may not understand their radiology reports [9-11]. For example, studies show approximately half of women with mammographic abnormalities did not understand that their results were abnormal despite the Mammography Quality Standards Act requirement that results are sent to the patient “in terms easily understood by a lay person” [11]. For patients to fully benefit from radiology report access, reports must be patient centered.

Several strategies for improving the radiology report patient experience have been proposed. For example, some authors have suggested a lay language summary at the end of reports [12,13], providing radiologist contact information in reports [14], or even offering immediate result consultation with a radiologist for certain examination types [15]. Others advocate for providing patient-centered reports for standardized examinations (eg, lung cancer screening reports) [16] or an annotated report with definitions and links to references and public-domain images [17].

The growing body of work in patient-centered radiology reporting shows significant progress toward patient-centered care. There are, however, a few shortcomings of the published literature. First, most of the work has been performed at large academic medical centers, and approximately 80% of radiologic studies are performed and interpreted in a private practice setting [18]. Second, much of the prior work has focused on a few specific report types. Third, most of the previous work requires additional time and effort on the part of interpreting radiologists. Therefore, we endeavored to pilot a commercially available software platform for delivering patient-centered radiology reports for all imaging modalities in an outpatient, private practice setting. Our purpose was to evaluate patient use and satisfaction with patient-centered radiology reports that were provided through a radiology-specific patient portal. We explored access and use characteristics across age groups, tracked engagement, and collected patient feedback.

METHODS

The study was conducted as a collaboration between three entities: (1) Diversified Radiology of Colorado (Lakewood, Colorado), a large subspecialty private practice radiology group; (2) Touchstone Medical Imaging, LLC (Plano, Texas), an operator of imaging centers across the United States; and (3) Scanslated, Inc (Tampa, Florida), a radiologist-founded start-up that has built software to convert the standard radiology report into a patient-centered interactive web page format (authors N.B., R.S., and J.K. are shareholders). From January 1, 2021, through May 31,

2021, all patients who had imaging performed at a group of seven Touchstone Medical Imaging outpatient imaging centers in the Denver, Colorado, metropolitan area and interpreted by a Diversified Radiology of Colorado radiologist were provided access to an online patient-centered radiology report via a radiology-specific patient portal. Patients were sent a text message link to their report 48 hours after a Diversified radiologist completed the report. Radiology reports were processed by the Scanslated software (Scanslated, Inc.; www.scanslated.com) and presented in an interactive web page-style format with diagrams and embedded plain language explanations of medical terms. The software analyzes radiology report text and uses natural language processing to identify and annotate medical terms and phrases. The annotated terms and phrases are then linked to content from a proprietary database of more than 7,000 plain language explanations. Patient-centered definitions and diagrams are viewable alongside the report within a self-contained interface (Fig. 1). Representative screenshots of the report access process are shown in Figure 2.

We recorded portal usage statistics, including the type of device (smartphone versus tablet or computer) used to access reports, report access rate, report view time, and the usage of interactive diagrams and plain language explanations. Results between patients who clicked versus those who did not click on the interactive features of the report (simple language definitions, diagrams) were compared. We assessed patients' subjective report comprehension and experience via two survey questions: (1) Did the definitions and diagrams help you to understand your report? (2) How was your overall experience of viewing your patient-centered report? For both questions, patients were given the option to select a positive, neutral, or negative response (Fig. 4). Results were stratified by age.

The Mann-Whitney *U* test was used to assess for differences in report viewing time and number of clicks between groups [19]. The χ^2 test was used to evaluate differences in categorical variables between groups. A significance level of .05 was used to determine statistical significance when comparing view time between patients who used the interactive plain language definitions and diagrams and patients who did not. For comparisons between different age groups, a Bonferroni correction was performed to adjust for multiple comparisons. Hence, $P \leq .002$ (.05/21) was considered statistically significant. Statistical testing was performed using the open-source software Scipy [20].

RESULTS

During the 5-month study period, a total of 16,735 individual patients viewed 21,679 unique radiology reports. Reports



Fig. 1. (A) Desktop and (B) mobile screenshots of an example patient-centered interactive radiology report. (C) Plain language explanations and diagrams appear when patients click medical terms and phrases.

consisted of all modalities (x-ray, CT, MR, ultrasound, nuclear medicine, mammography) and examination types performed at the outpatient imaging centers included in the study. Overall, 59.2% (16,735 of 28,276) of patients with an available report accessed the report in the online portal. The average number of views per report was 1.8 (38,547 total report views; 21,679 unique viewed reports). In all, 87.7% (33,858 of 38,547) of report views occurred on mobile devices, and 59.9% (12,993 of 21,679) of unique patients interacted with the report by clicking on terms and diagrams. For patients who used the interactive plain language definitions and diagrams, the average report viewing time was 5.8 ± 6.6 min, compared with 1.8 ± 1.9 min for patients who chose to only view the raw text report ($P < .005$) (Fig. 3). In addition, 84.7% (3,847 of 4,544) of patients reported that the definitions and diagrams helped them to understand their report, and 86.7% (3,296 of 3,801) of patients

endorsed an overall positive experience viewing their report online (Fig. 4).

Report view time and total number of clicks for each of 21,679 unique report views were stratified into seven groups by patient age (<18, 18-29, 30-39, 40-49, 50-59, 60-69, and >70). There was no significant difference in report access rate between age groups 18 to 29, 30 to 39, 40 to 49, 50 to 59, and 60 to 69 (range, 59%-64% access). Report access rate was lower in the <18-year-old age group and in the >70-year-old age group (40.5% and 48.9% access rate, respectively). Report view time was significantly greater in the >70-year-old age group (4.9 min per report) compared with the other age groups (range 2.9 min to 4.5 min, all $P < .002$). Report interaction was significantly greater in the 18- to 29-year-old age group (mean of 7.8 clicks on interactive definitions and diagrams per report) compared with the <18, 40 to 49, 50 to 59, 60 to 69, and >70 age groups (range, 4.5-

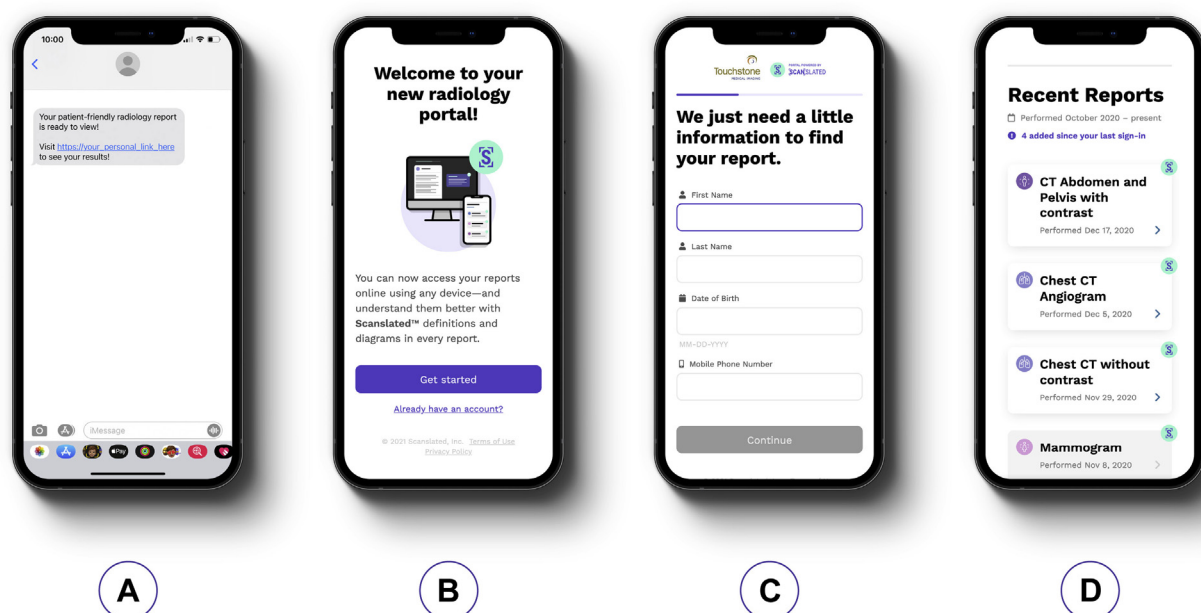


Fig. 2. (A) Patients were sent a text message with a unique URL providing access to their radiology report. (B) Welcome screen for radiology-specific patient portal. (C) Patient identity was verified with first and last name, date of birth, telephone number, and a two-factor authentication step. (D) Once authenticated, patients could view their available radiology reports in chronological order. Clicking any report icon generates a web page-style report with diagrams and embedded plain language explanations of medical terms.

6.9 clicks per report, all $P < .002$). There was no significant difference in report engagement between the 18 to 29 and the 30 to 39 age groups (7.8 versus 6.5 clicks per report, $P = .009$). Reports accessed by patients in the >70-year-old group were more likely to be viewed on a computer or tablet (17% versus 4%-13% for other age groups, $\chi^2 [6, n = 21,679] = 411.3, P < .0001$). These data are summarized in Table 1.

DISCUSSION

The direct and immediate availability of radiology reports to patients represents both a challenge and an opportunity for radiologists. Access to electronic health records is intended to empower patients; however, the traditional radiology report is difficult for patients to understand and use [9-11]. There is a growing realization among radiologists that a patient-centered report could add significant value to our

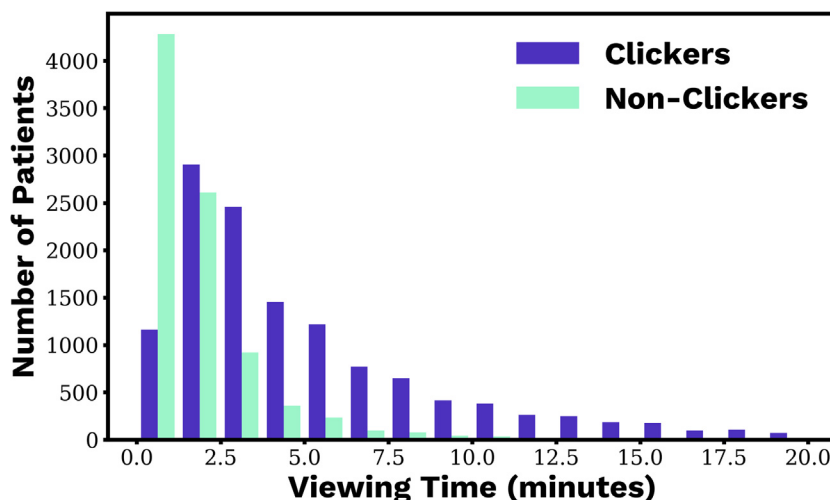


Fig. 3. Comparison of report view time for patients who used the interactive plain language definitions and diagrams (clickers) vs viewing the raw text report only (nonclickers) (mean 5.8 ± 6.6 min vs 1.8 ± 1.9 min).

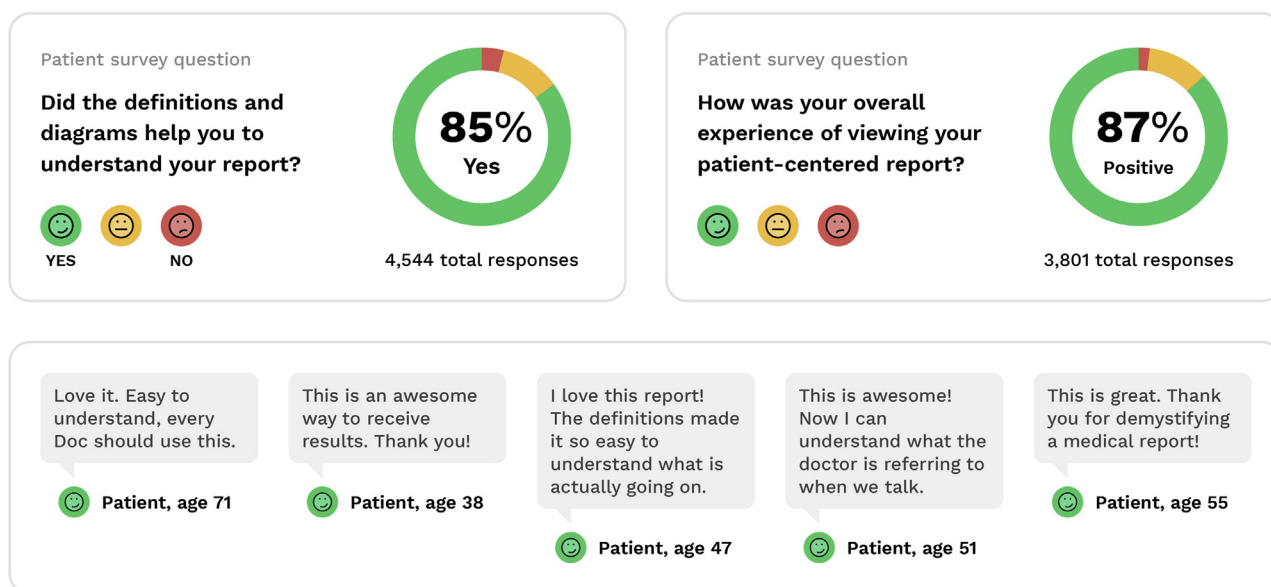


Fig. 4. Patients were presented with feedback questions after viewing the patient-centered interactive report and were invited to leave a comment. In all, 84.7% (3,847 of 4,544) of patients reported that definitions and diagrams helped them understand their report. In addition, 86.7% (3,296 of 3,801) of patients endorsed an overall positive experience viewing their report online.

work product, increase the visibility of radiologists, and empower patients to be more engaged in their care [21,22]. In the present study, we demonstrate the feasibility and potential of patient-centered radiology reports in the outpatient private practice setting. We piloted a commercially available software platform for delivering patient-centered radiology reports across all imaging modalities. Importantly, the patient-centric report does not alter the wording of the original report but rather contains embedded plain language explanations and diagrams intended to present the original report in a format that will be more

accessible and useful to patients. We found that most patients (59%) with an available report used the portal to view their report online. We also found that dedicated patient-centered content may increase patient engagement, as the average report viewing time was 4 min longer for patients who used the interactive plain language definitions and diagrams.

The percentage of patients accessing their radiology reports in the present study (59%) compares favorably to prior studies. For comparison, Miles et al showed a report access rate of 51% in their large patient cohort [4]. In a smaller

Table 1. Online patient-centered report access and engagement across various age groups

Age Group (y)	Patient Access Rate	Reports Viewed	Report View Time (Min)	Number of Clicks
<18	40.5% (214 of 528)	248	2.9	4.5
18-29	63.7% (1,400 of 2,198)	1,732	3.3	7.8
30-39	63.6% (2,162 of 3,398)	2,733	3.5	6.5
40-49	63.0% (2,692 of 4,271)	3,435	3.9	6.6
50-59	63.3% (3,592 of 5,678)	4,671	4.4	6.9
60-69	59.4% (4,010 of 6,756)	5,270	4.5	7
>70	48.9% (2,665 of 5,447)	3,590	4.9	6.7

pilot of annotated knee MRI reports, Cook et al showed a report access rate of 16.3% [17]. Increased report access rate in the present study may be multifactorial in etiology. First, the prior studies took place approximately 4 to 5 years ago, and baseline patient portal use has increased over that time. Other potential explanations include the direct engagement of patients via text message, the use of a radiology-specific patient portal, and the availability of patient-centered content.

Patients <18 or >70 years of age were less likely to access their online report than patients in the stratified 18 to 69 age groups. In the >70-year-old age group, these differences may be due to decreased comfort and familiarity with the digital account creation and authentication process. In the <18-year-old age group, many users likely represent parents logging in to receive results on behalf of their children. It is unclear why there was a lower report access rate in this demographic, although this finding may in part be related to small sample size ($n = 248$ patients in the <18 age group compared with $n > 1,000$ for all other age groups).

Time spent reading and interacting with a radiology report may be considered a measure of patient engagement [9]. In our study, patients using the patient-centered content spent significantly more time viewing their reports. This finding is consistent with prior data, which showed that patients desire lay language explanations of their report results [23]. As assessed by use of interactive diagrams and plain language definitions, report engagement was overall greatest in the 18- to 29-year-old age group, which may be more accustomed to interacting with web-based platforms. Interestingly, patients >70 years old spent more time viewing their online reports than other age groups. These findings suggest that patient-centered content may increase engagement and that online patient-centered radiology reports may represent a feasible patient-engagement strategy for patients of all ages.

Our early experience suggests that patient-centered radiology reports may improve subjective patient comprehension and positively impact the patient experience. A majority of patients reported that the plain language definitions and diagrams helped them to understand their report. In addition, 86.7% endorsed an overall positive experience viewing their report online in this format. Furthermore, patient feedback collected during the study period was overwhelmingly positive, suggesting that patients see value in online report access and patient-centered content.

Our study has several limitations. Although patients did report a subjective increase in perceived report comprehension, we did not implement a formal objective measure of comprehension to confirm this. Because patients were not randomized to interactive plain language definitions and

diagrams versus raw text report only, the differences in average viewing time could be attributable to factors other than patient-centered radiology reporting. We did not survey referring physicians' opinions on the patient-centered content, track the number of calls to referring physician offices, or assess the impact on referral patterns. In the future, we plan to conduct user experience interviews to gain feedback on how the patient-centric reports would impact patients' overall ability to engage in their health care. Our hope is that an improved understanding of radiology results may better prepare and empower patients to meaningfully participate in follow-up visits with their doctors. Furthermore, well-informed patients may have fewer questions for their ordering providers in advance of scheduled follow-up appointments, which could decrease time spent answering imaging-related telephone calls and e-mail questions and in turn increase the proportion of clinician time dedicated to direct patient care.

In conclusion, our results suggest that patient-centered reporting is feasible in the outpatient setting and that dedicated patient-centered content has the potential to increase engagement across all age groups. To our knowledge, this is the first implementation of online patient-centered radiology reporting for all report types in a private practice outpatient setting.

Future work should investigate factors that influence differences in radiology report access rates and use across age groups to inform strategies to optimize the radiology report viewing experience for all patients. Additionally, further study is needed on which report types are of greatest interest to patients and on the downstream impact of patient-centered reports on physician time, patient follow-up adherence, and ultimately health outcomes.

TAKE-HOME POINTS

- We piloted a commercially available software platform for delivering patient-centered radiology reports across all imaging modalities in a private practice setting.
- We found that 59% of patients accessed their radiology report via our radiology-specific patient portal, higher than previously published.
- Average report viewing time was 4 min longer for patients who used the interactive plain language definitions and diagrams, suggesting that dedicated patient-centered content may increase engagement.
- Of the patients viewing their reports, 84.7% reported that the plain language definitions and diagrams helped them to understand their report, and 86.7% endorsed an overall positive experience viewing their report online in this format.

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